

DS311 - R Lab Assignment

Shaun Kumar

2025-04-27

R Assignment 1

- In this assignment, we are going to apply some of the built-in data set in R for descriptive statistics analysis.
- To earn full grade in this assignment, students need to complete the coding tasks for each question to get the result.
- After finished all the questions, knit the document into HTML format for submission.

Question 1

Using the **mtcars** data set in R, please answer the following questions.

```
# Loading the data
data(mtcars)

# Head of the data set
head(mtcars)
```

	mpg	cyl	dis	hp	drat	wt	qsec	vs	am
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1
Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0
Hornet Sportabout	18.7	8	360	175	3.15	3.440	17.02	0	0
Valiant	18.1	6	225	105	2.76	3.460	20.22	1	0

6 rows | 1-10 of 12 columns

a. Report the number of variables and observations in the data set.

```
# Enter your code here!
num_obs <- nrow(mtcars)
num_vars <- ncol(mtcars)

# Answer:
print(paste("There are total of", num_vars, "variables and", num_obs, "observations in this data set."))
```

```
## [1] "There are total of 11 variables and 32 observations in this data set."
```

b. Print the summary statistics of the data set and report how many discrete and continuous variables are in the data set.

```
# Enter your code here!
summary(mtcars)
```

```
##      mpg      cyl      disp      hp
##  Min.   :10.40  Min.   :4.000  Min.   : 71.1  Min.   : 52.0
## 1st Qu.:15.43 1st Qu.:4.000 1st Qu.:120.8 1st Qu.: 96.5
## Median :19.20 Median :6.000 Median :196.3 Median :123.0
## Mean   :20.09 Mean   :6.188 Mean   :230.7 Mean   :146.7
## 3rd Qu.:22.80 3rd Qu.:8.000 3rd Qu.:326.0 3rd Qu.:180.0
## Max.   :33.90 Max.   :8.000 Max.   :472.0 Max.   :335.0
##      drat      wt      qsec      vs
##  Min.   :2.760  Min.   :1.513  Min.   :14.50  Min.   :0.0000
## 1st Qu.:3.080 1st Qu.:2.581 1st Qu.:16.89 1st Qu.:0.0000
## Median :3.695 Median :3.325 Median :17.71 Median :0.0000
## Mean   :3.597 Mean   :3.217 Mean   :17.85 Mean   :0.4375
## 3rd Qu.:3.920 3rd Qu.:3.610 3rd Qu.:18.90 3rd Qu.:1.0000
## Max.   :4.930 Max.   :5.424 Max.   :22.90 Max.   :1.0000
##      am      gear      carb
##  Min.   :0.0000  Min.   :3.000  Min.   :1.000
## 1st Qu.:0.0000 1st Qu.:3.000 1st Qu.:2.000
## Median :0.0000 Median :4.000 Median :2.000
## Mean   :0.4062 Mean   :3.688 Mean   :2.812
## 3rd Qu.:1.0000 3rd Qu.:4.000 3rd Qu.:4.000
## Max.   :1.0000 Max.   :5.000 Max.   :8.000
```

```
discrete_vars <- sapply(mtcars, function(x) is.integer(x) || (is.numeric(x) && length(unique(x)) < 10))
continuous_vars <- !discrete_vars

num_discrete <- sum(discrete_vars)
num_continuous <- sum(continuous_vars)

# Answer:
print(paste("There are", num_discrete, "discrete variables and", num_continuous, "continuous variables in this data set."))
```

```
## [1] "There are 5 discrete variables and 6 continuous variables in this data set."
```

c. Calculate the mean, variance, and standard deviation for the variable **mpg** and assign them into variable names **m**, **v**, and **s**. Report the results in the print statement.

```
# Enter your code here!
m <- round(mean(mtcars$mpg),2)
v <- round(var(mtcars$mpg),2)
s <- round(sd(mtcars$mpg),2)

print(paste("The average of Mile Per Gallon from this data set is", m , "with variance", v , "and standard deviation", s, "."))
```

```
## [1] "The average of Mile Per Gallon from this data set is 20.09 with variance 36.32 and standard deviation 6.03 ."
```

d. Create two tables to summarize 1) average mpg for each cylinder class and 2) the standard deviation of mpg for each gear class.

```
# Enter your code here!
avg_mpg_by_cyl <- aggregate(mpg ~ cyl, data = mtcars, FUN = mean)
sd_mpg_by_gear <- aggregate(mpg ~ gear, data = mtcars, FUN = sd)

print(avg_mpg_by_cyl)
```

```
##   cyl      mpg
## 1    4 26.66364
## 2    6 19.74286
## 3    8 15.10000
```

```
print(sd_mpg_by_gear)
```

```
##   gear      mpg
## 1    3 3.371618
## 2    4 5.276764
## 3    5 6.658979
```

e. Create a crosstab that shows the number of observations belong to each cylinder and gear class combinations. The table should show how many observations given the car has 4 cylinders with 3 gears, 4 cylinders with 4 gears, etc. Report which combination is recorded in this data set and how many observations for this type of car.

```
# Enter your code here!
crosstab <- table(mtcars$cyl, mtcars$gear)
max_count <- max(crosstab)
max_position <- which(crosstab == max_count, arr.ind = TRUE)

most_common_cyl <- as.numeric(rownames(crosstab)[max_position[1]])
most_common_gear <- as.numeric(colnames(crosstab)[max_position[2]])

print(paste("The most common car type in this data set is car with",most_common_cyl,"cylinders and",most_common_gear,"gears. There are total of",max_count,"cars belong to this specification in the data set."))
```

```
## [1] "The most common car type in this data set is car with 8 cylinders and 3 gears. There are total of 12 cars belong to this specification in the data set."
```

Question 2

Use different visualization tools to summarize the data sets in this question.

- Using the **PlantGrowth** data set, visualize and compare the weight of the plant in the three separated group. Give labels to the title, x-axis, and y-axis on the graph. Write a paragraph to summarize your findings.

```
# Load the data set
data("PlantGrowth")

# Head of the data set
head(PlantGrowth)
```

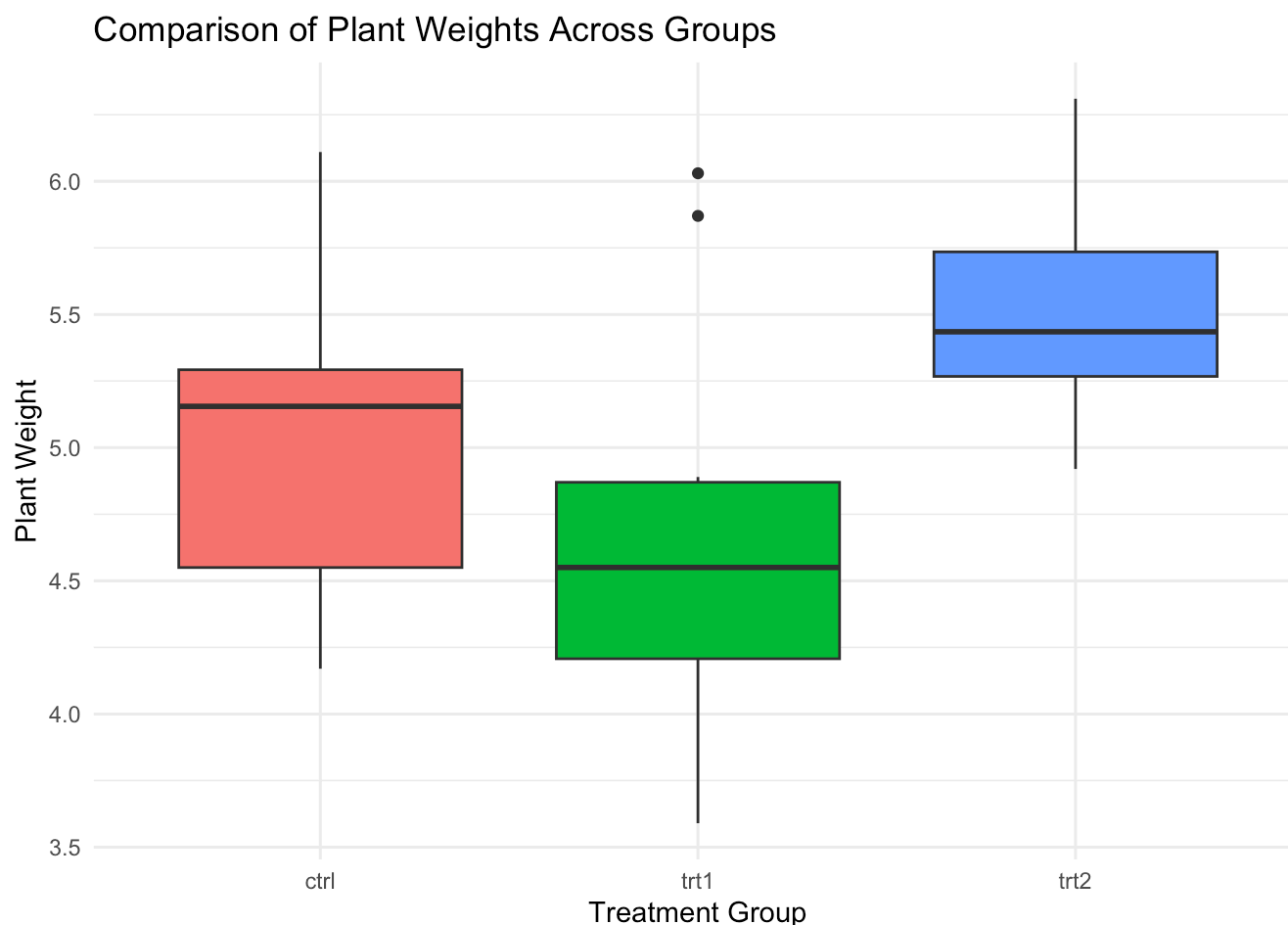
	weight	group
	<dbl>	<fct>
1	4.17	ctrl
2	5.58	ctrl
3	5.18	ctrl
4	6.11	ctrl
5	4.50	ctrl
6	4.61	ctrl

6 rows

```
# Enter your code here!
# Set a CRAN mirror before installing packages
options(repos = c(CRAN = "https://cran.rstudio.com"))
install.packages("ggplot2")
```

```
##  
## The downloaded binary packages are in  
## /var/folders/bb/qxcf7b0d1zzbx6vt008mk2r00000gn/T//Rtmp40ZGN4/downloaded_packages
```

```
library(ggplot2)  
  
ggplot(PlantGrowth, aes(x = group, y = weight, fill = group)) +  
  geom_boxplot() +  
  labs(  
    title = "Comparison of Plant Weights Across Groups",  
    x = "Treatment Group",  
    y = "Plant Weight"  
  ) +  
  theme_minimal() +  
  theme(legend.position = "none")
```

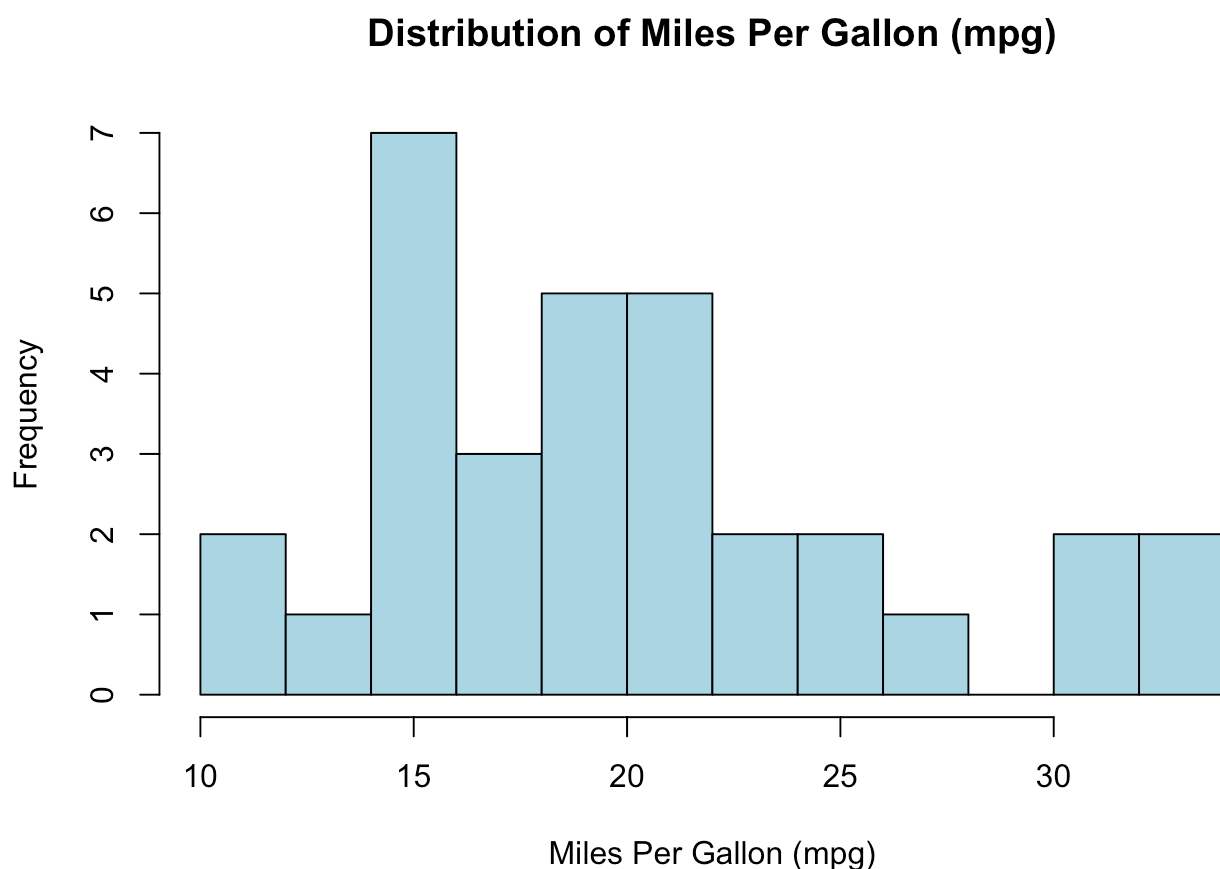


Result:

=> Report a paragraph to summarize your findings from the plot! The boxplot shows how plant weights differ between the three groups: control, treatment 1 (trt1), and treatment 2 (trt2). The control group has a fairly steady range of plant weights, with most plants weighing around 5. Treatment 1 seems to result in slightly lower plant weights, while treatment 2 shows slightly higher weights compared to the control group. However, the ranges of weights for all three groups overlap, so it's not clear if the differences are meaningful just by looking at the plot.

- b. Using the **mtcars** data set, plot the histogram for the column **mpg** with 10 breaks. Give labels to the title, x-axis, and y-axis on the graph. Report the most observed mpg class from the data set.

```
hist_result <- hist(  
  mtcars$mpg,  
  breaks = 10,  
  main = "Distribution of Miles Per Gallon (mpg)",  
  xlab = "Miles Per Gallon (mpg)",  
  ylab = "Frequency",  
  col = "lightblue",  
  border = "black"  
)
```



```
max_bin_index <- which.max(hist_result$counts)  
most_common_range <- paste(round(hist_result$breaks[max_bin_index], 1),  
  "-",  
  round(hist_result$breaks[max_bin_index + 1], 1))  
  
print(paste("Most of the cars in this data set are in the class of", most_common_range, "m  
  iles per gallon."))
```

```
## [1] "Most of the cars in this data set are in the class of 14 - 16 miles per gallon."
```

- c. Using the **USArrests** data set, create a pairs plot to display the correlations between the variables in the data set. Plot the scatter plot with **Murder** and **Assault**. Give labels to the title, x-axis, and y-axis on the graph. Write a paragraph to summarize your results from both plots.

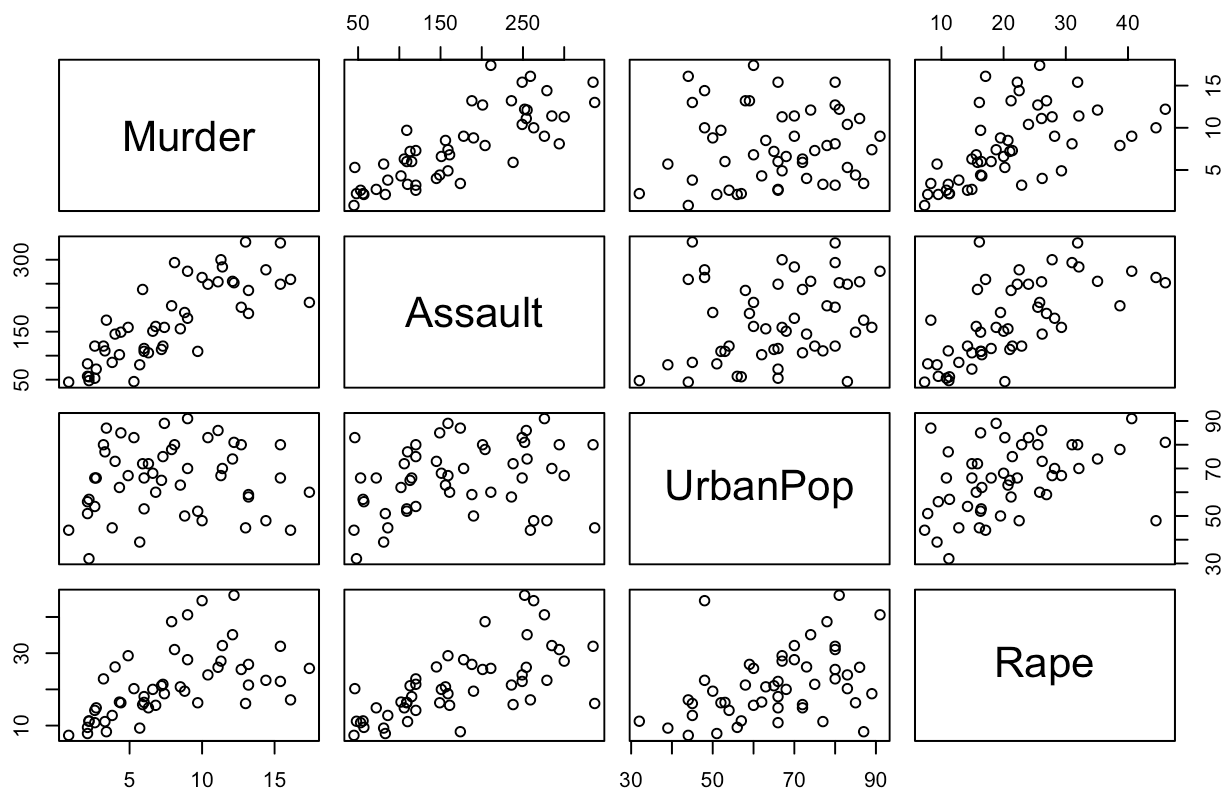
```
# Load the data set
data("USArrests")

# Head of the data set
head(USArrests)
```

	Murder <dbl>	Assault <int>	UrbanPop <int>	Rape <dbl>
Alabama	13.2	236	58	21.2
Alaska	10.0	263	48	44.5
Arizona	8.1	294	80	31.0
Arkansas	8.8	190	50	19.5
California	9.0	276	91	40.6
Colorado	7.9	204	78	38.7
6 rows				

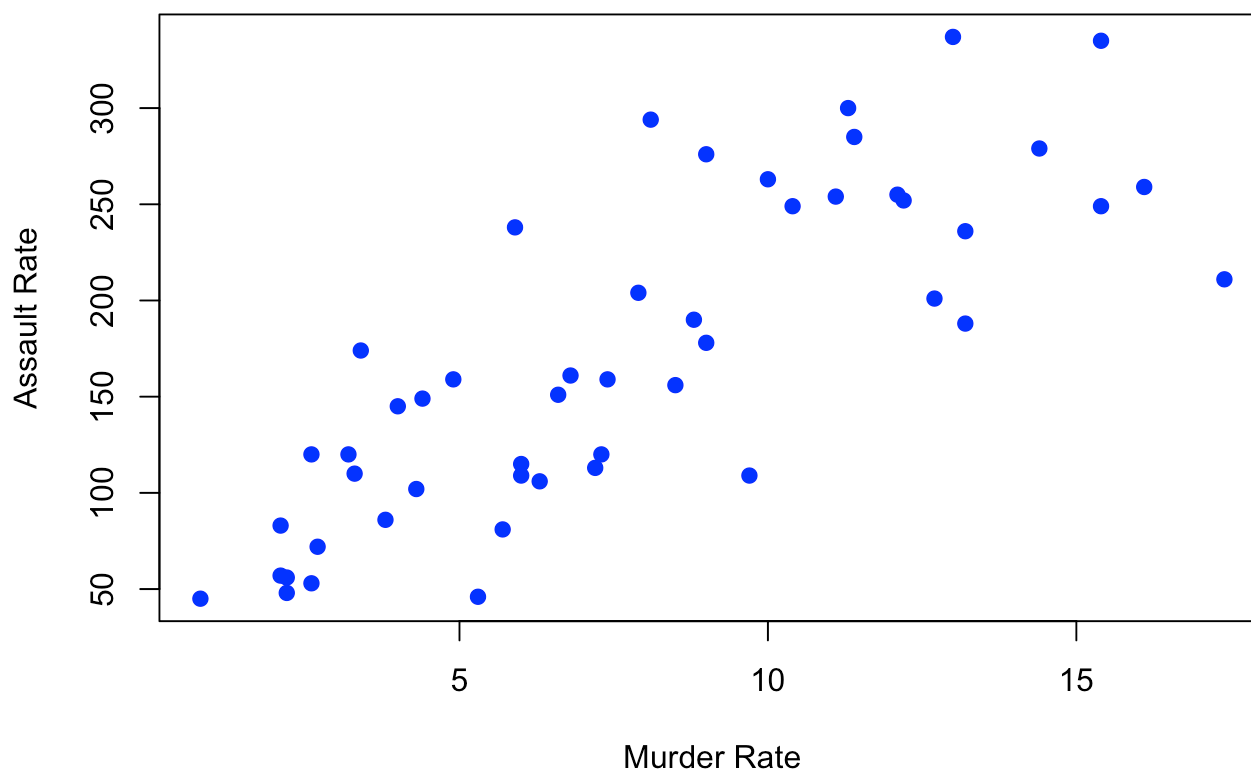
```
# Enter your code here!
pairs(USArrests, main = "Pairs Plot of USArrests Data")
```

Pairs Plot of USArrests Data



```
plot(USArrests$Murder, USArrests$Assault,
     main = "Scatter Plot of Murder vs Assault",
     xlab = "Murder Rate",
     ylab = "Assault Rate",
     col = "blue",
     pch = 19) # pch=19 gives solid circles
```

Scatter Plot of Murder vs Assault



Result:

=> Report a paragraph to summarize your findings from the plot!

The plot shows the relationship between the different crime rates in the USArrests dataset. It highlights a clear positive correlation between Murder and Assault, where states with higher murder rates tend to have higher assault rates as well. The scatter plot further confirms this, showing an upward trend between these two variables. Overall, the plots suggest that murder and assault are closely related.

Question 3

Download the housing data set from www.jaredlander.com and find out what explains the housing prices in New York City.

Note: Check your working directory to make sure that you can download the data into the data folder.

- Create your own descriptive statistics and aggregation tables to summarize the data set and find any meaningful results between different variables in the data set.

```
# Head of the cleaned data set  
head(housingData)
```

Neighborhood <chr>	Market.Value.per.SqFt <dbl>	Boro <chr>	Year.Built <int>
1 FINANCIAL	200.00	Manhattan	1920
2 FINANCIAL	242.76	Manhattan	1985
4 FINANCIAL	271.23	Manhattan	1930
5 TRIBECA	247.48	Manhattan	1985
6 TRIBECA	191.37	Manhattan	1986
7 TRIBECA	211.53	Manhattan	1985
6 rows			

Enter your code here!
summary(housingData)

```
## Neighborhood      Market.Value.per.SqFt      Boro      Year.Built
## Length:2530      Min.   : 10.66      Length:2530      Min.   :1825
## Class :character  1st Qu.: 75.10      Class :character  1st Qu.:1926
## Mode  :character  Median :114.89      Mode  :character  Median :1986
##                  Mean   :133.17      Mean   :1967
##                  3rd Qu.:189.91      3rd Qu.:2005
##                  Max.    :399.38      Max.    :2010
```

aggregate(Market.Value.per.SqFt ~ Boro, data = housingData, FUN = mean)

Boro <chr>	Market.Value.per.SqFt <dbl>
Bronx	47.93232
Brooklyn	80.13439
Manhattan	180.59265
Queens	77.38137
Staten Island	41.26958
5 rows	

aggregate(Year.Built ~ Boro, data = housingData, FUN = mean)

Boro <chr>	Year.Built <dbl>
Bronx	1983.841

Boro <chr>	Year.Built <dbl>
Brooklyn	1983.906
Manhattan	1953.803
Queens	1983.866
Staten Island	1980.833
5 rows	

```
cor(housingData$Market.Value.per.SqFt, housingData$Year.Built)
```

```
## [1] -0.09559073
```

```
table(housingData$Boro, housingData$Neighborhood)
```

##							
##		ALPHABET CITY	ARROCHAR-SHORE	ACRES	ASTORIA	BATH BEACH	BAY RIDGE
##	Bronx	0		0	0	0	0
##	Brooklyn	0		0	0	6	8
##	Manhattan	24		0	0	0	0
##	Queens	0		0	30	0	0
##	Staten Island	0		1	0	0	0
##							
##		BAYSIDE	BEDFORD PARK/NORWOOD	BEDFORD	STUYVESANT	BELMONT	
##	Bronx	0		2		0	1
##	Brooklyn	0		0		58	0
##	Manhattan	0		0		0	0
##	Queens	9		0		0	0
##	Staten Island	0		0		0	0
##							
##		BENSONHURST	BERGEN BEACH	BOERUM HILL	BOROUGH PARK	BRIARWOOD	
##	Bronx	0	0	0		0	0
##	Brooklyn	7	1	15		35	0
##	Manhattan	0	0	0		0	0
##	Queens	0	0	0		0	4
##	Staten Island	0	0	0		0	0
##							
##		BRIGHTON BEACH	BRONX-UNKNOWN	BRONXDALE	BROOKLYN HEIGHTS		
##	Bronx	0		2	3		0
##	Brooklyn	21		0	0		9
##	Manhattan	0		0	0		0
##	Queens	0		0	0		0
##	Staten Island	0		0	0		0
##							
##		BUSH TERMINAL	BUSHWICK	CANARSIE	CARROLL GARDENS	CHELSEA	
##	Bronx	0	0	0		0	0
##	Brooklyn	1	2	4		9	0
##	Manhattan	0	0	0		0	88
##	Queens	0	0	0		0	0
##	Staten Island	0	0	0		0	0
##							
##		CHINATOWN	CITY ISLAND	CIVIC CENTER	CLINTON	CLINTON HILL	
##	Bronx	0	1	0	0		0
##	Brooklyn	0	0	0	0		13
##	Manhattan	21	0	23	31		0
##	Queens	0	0	0	0		0
##	Staten Island	0	0	0	0		0
##							
##		COBBLE HILL	COBBLE HILL-WEST	COLLEGE POINT	CONEY ISLAND	CORONA	
##	Bronx	0	0		0	0	0
##	Brooklyn	5	8		0	4	0
##	Manhattan	0	0		0	0	0
##	Queens	0	0		3	0	17
##	Staten Island	0	0		0	0	0
##							
##		CROWN HEIGHTS	DOWNTOWN-FULTON	FERRY DOWNTOWN-FULTON	MALL		
##	Bronx	0		0		0	

##	Brooklyn	14		14		2
##	Manhattan	0		0		0
##	Queens	0		0		0
##	Staten Island	0		0		0
##						
##		DOWNTOWN-METROTECH DYKER HEIGHTS EAST NEW YORK EAST TREMONT				
##	Bronx	0		0		3
##	Brooklyn	3		1		0
##	Manhattan	0		0		0
##	Queens	0		0		0
##	Staten Island	0		0		0
##						
##		EAST VILLAGE ELMHURST FAR ROCKAWAY FASHION FINANCIAL				
##	Bronx	0	0	0	0	0
##	Brooklyn	0	0	0	0	0
##	Manhattan	26	0	0	15	36
##	Queens	0	55	4	0	0
##	Staten Island	0	0	0	0	0
##						
##		FLATBUSH-CENTRAL FLATBUSH-LEFFERTS GARDEN FLATBUSH-NORTH				
##	Bronx	0			0	0
##	Brooklyn	6			1	8
##	Manhattan	0			0	0
##	Queens	0			0	0
##	Staten Island	0			0	0
##						
##		FLATIRON FLUSHING MEADOW PARK FLUSHING-NORTH FLUSHING-SOUTH				
##	Bronx	0		0		0
##	Brooklyn	0		0		0
##	Manhattan	45		0		0
##	Queens	0		1	133	8
##	Staten Island	0		0		0
##						
##		FOREST HILLS FORT GREENE GLENDALE GOWANUS GRAMERCY GRANT CITY				
##	Bronx	0	0	0	0	0
##	Brooklyn	0	10	0	3	0
##	Manhattan	0	0	0	0	17
##	Queens	17	0	3	0	0
##	Staten Island	0	0	0	0	2
##						
##		GRAVESEND GREAT KILLS GREENPOINT GREENWICH VILLAGE-CENTRAL				
##	Bronx	0	0	0		0
##	Brooklyn	19	0	19		0
##	Manhattan	0	0	0		60
##	Queens	0	0	0		0
##	Staten Island	0	1	0		0
##						
##		GREENWICH VILLAGE-WEST GRYMES HILL HAMMELS HARLEM-CENTRAL				
##	Bronx		0	0	0	0
##	Brooklyn		0	0	0	0
##	Manhattan		57	0	0	94
##	Queens		0	0	5	0

##	Staten Island	0	1	0	0		
##							
##		HARLEM-EAST	HARLEM-UPPER	HARLEM-WEST	HIGHBRIDGE/MORRIS HEIGHTS		
##	Bronx	0	0	0	1		
##	Brooklyn	0	0	0	0		
##	Manhattan	36	12	2	0		
##	Queens	0	0	0	0		
##	Staten Island	0	0	0	0		
##							
##		HILLCREST	HOLLIS	HOWARD BEACH	INWOOD JACKSON HEIGHTS JAMAICA		
##	Bronx	0	0	0	0	0	
##	Brooklyn	0	0	0	0	0	
##	Manhattan	0	0	0	2	0	0
##	Queens	1	1	5	0	21	5
##	Staten Island	0	0	0	0	0	0
##							
##		JAMAICA ESTATES	JAVITS CENTER	KENSINGTON	KEW GARDENS		
##	Bronx	0	0	0	0		
##	Brooklyn	0	0	2	0		
##	Manhattan	0	1	0	0		
##	Queens	2	0	0	10		
##	Staten Island	0	0	0	0		
##							
##		KINGSBRIDGE HTS/UNIV HTS	KINGSBRIDGE/JEROME PARK	KIPS BAY			
##	Bronx	1	5	0			
##	Brooklyn	0	0	0			
##	Manhattan	0	0	13			
##	Queens	0	0	0			
##	Staten Island	0	0	0			
##							
##		LITTLE ITALY	LITTLE NECK	LONG ISLAND CITY	LOWER EAST SIDE		
##	Bronx	0	0	0	0		
##	Brooklyn	0	0	0	0		
##	Manhattan	13	0	0	42		
##	Queens	0	1	24	0		
##	Staten Island	0	0	0	0		
##							
##		MADISON MANHATTAN VALLEY	MASPETH	MIDDLE VILLAGE	MIDTOWN CBD		
##	Bronx	0	0	0	0	0	
##	Brooklyn	19	0	0	0	0	
##	Manhattan	0	23	0	0	13	
##	Queens	0	0	4	7	0	
##	Staten Island	0	0	0	0	0	
##							
##		MIDTOWN EAST	MIDTOWN WEST	MIDWOOD	MORNINGSIDE	HEIGHTS	
##	Bronx	0	0	0	0	0	
##	Brooklyn	0	0	11	0	0	
##	Manhattan	56	47	0	1	0	
##	Queens	0	0	0	0	0	
##	Staten Island	0	0	0	0	0	
##							
##		MORRIS PARK/VAN NEST	MORRISANIA/LONGWOOD	MOTT HAVEN/PORT MORRIS			

##	Bronx	1	16	1
##	Brooklyn	0	0	0
##	Manhattan	0	0	0
##	Queens	0	0	0
##	Staten Island	0	0	0
##				
##	MURRAY HILL NEW BRIGHTON NEW BRIGHTON-ST. GEORGE			
##	Bronx	0	0	0
##	Brooklyn	0	0	0
##	Manhattan	39	0	0
##	Queens	0	0	0
##	Staten Island	0	3	2
##				
##	NEW SPRINGVILLE OAKLAND GARDENS OCEAN HILL OCEAN PARKWAY-NORTH			
##	Bronx	0	0	0
##	Brooklyn	0	0	10
##	Manhattan	0	0	0
##	Queens	0	1	0
##	Staten Island	9	0	0
##				
##	OCEAN PARKWAY-SOUTH OZONE PARK PARK SLOPE PARK SLOPE SOUTH			
##	Bronx	0	0	0
##	Brooklyn	1	0	31
##	Manhattan	0	0	0
##	Queens	0	1	0
##	Staten Island	0	0	0
##				
##	PARKCHESTER PELHAM PARKWAY SOUTH PROSPECT HEIGHTS REGO PARK			
##	Bronx	2	1	0
##	Brooklyn	0	0	5
##	Manhattan	0	0	0
##	Queens	0	0	0
##	Staten Island	0	0	27
##				
##	RIDGEWOOD RIVERDALE ROCKAWAY PARK SCHUYLerville/PELHAM BAY			
##	Bronx	0	17	0
##	Brooklyn	0	0	0
##	Manhattan	0	0	0
##	Queens	3	0	5
##	Staten Island	0	0	0
##				
##	SHEEPSHEAD BAY SILVER LAKE SOHO SOUNDVIEW SOUTH OZONE PARK			
##	Bronx	0	0	0
##	Brooklyn	27	0	0
##	Manhattan	0	0	55
##	Queens	0	0	0
##	Staten Island	0	2	0
##				
##	SOUTHBRIDGE SUNNYSIDE SUNSET PARK THROGS NECK TOMPKINSVILLE			
##	Bronx	0	0	0
##	Brooklyn	0	0	23
##	Manhattan	9	0	0

##	Queens	0	10	0	0	0
##	Staten Island	0	1	0	0	1
##						
##		TRIBECA UPPER EAST SIDE (59-79)		UPPER EAST SIDE (79-96)		
##	Bronx	0		0		0
##	Brooklyn	0		0		0
##	Manhattan	74		123		78
##	Queens	0		0		0
##	Staten Island	0		0		0
##						
##		UPPER EAST SIDE (96-110)		UPPER WEST SIDE (59-79)		
##	Bronx		0		0	
##	Brooklyn		0		0	
##	Manhattan		5		87	
##	Queens		0		0	
##	Staten Island		0		0	
##						
##		UPPER WEST SIDE (79-96)		UPPER WEST SIDE (96-116)		
##	Bronx		0		0	
##	Brooklyn		0		0	
##	Manhattan		66		34	
##	Queens		0		0	
##	Staten Island		0		0	
##						
##		WASHINGTON HEIGHTS LOWER WASHINGTON HEIGHTS		UPPER		
##	Bronx		0		0	
##	Brooklyn		0		0	
##	Manhattan		5		6	
##	Queens		0		0	
##	Staten Island		0		0	
##						
##		WEST NEW BRIGHTON WHITESTONE		WILLIAMSBRIDGE		
##	Bronx	0	0		2	
##	Brooklyn	0	0		0	
##	Manhattan	0	0		0	
##	Queens	0	2		0	
##	Staten Island	1	0		0	
##						
##		WILLIAMSBURG-CENTRAL		WILLIAMSBURG-EAST		WILLIAMSBURG-NORTH
##	Bronx	0		0		0
##	Brooklyn	60		38		26
##	Manhattan	0		0		0
##	Queens	0		0		0
##	Staten Island	0		0		0
##						
##		WILLIAMSBURG-SOUTH WINDSOR TERRACE		WOODHAVEN WOODSIDE		
##	Bronx	0		0	0	0
##	Brooklyn	34		5	0	0
##	Manhattan	0		0	0	0
##	Queens	0		0	4	8
##	Staten Island	0		0	0	0
##						

```
##                WYCKOFF HEIGHTS
##   Bronx                0
##   Brooklyn            3
##   Manhattan           0
##   Queens              0
##   Staten Island       0
```

```
install.packages("dplyr")
```

```
##
## The downloaded binary packages are in
## /var/folders/bb/qxcf7b0d1zzbx6vt008mk2r00000gn/T//Rtmp40ZGN4/downloaded_packages
```

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##   filter, lag
```

```
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
housingData %>%
  group_by(Boro) %>%
  summarise(
    Avg_Value = mean(Market.Value.per.SqFt),
    Avg_Year = mean(Year.Built),
    Count = n()
  )
```

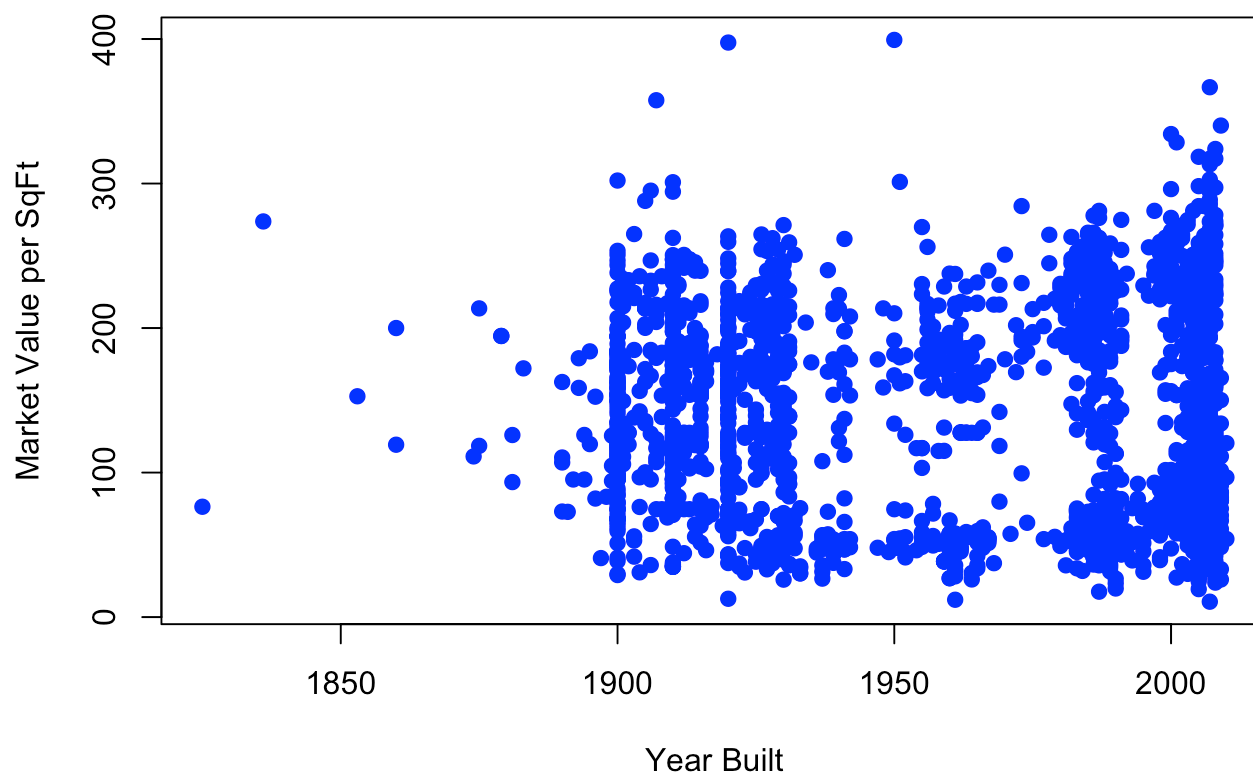
Boro <chr>	Avg_Value <dbl>	Avg_Year <dbl>	Count <int>
Bronx	47.93232	1983.841	69
Brooklyn	80.13439	1983.906	626
Manhattan	180.59265	1953.803	1379
Queens	77.38137	1983.866	432
Staten Island	41.26958	1980.833	24

5 rows

- b. Create multiple plots to demonstrate the correlations between different variables. Remember to label all axes and give title to each graph.

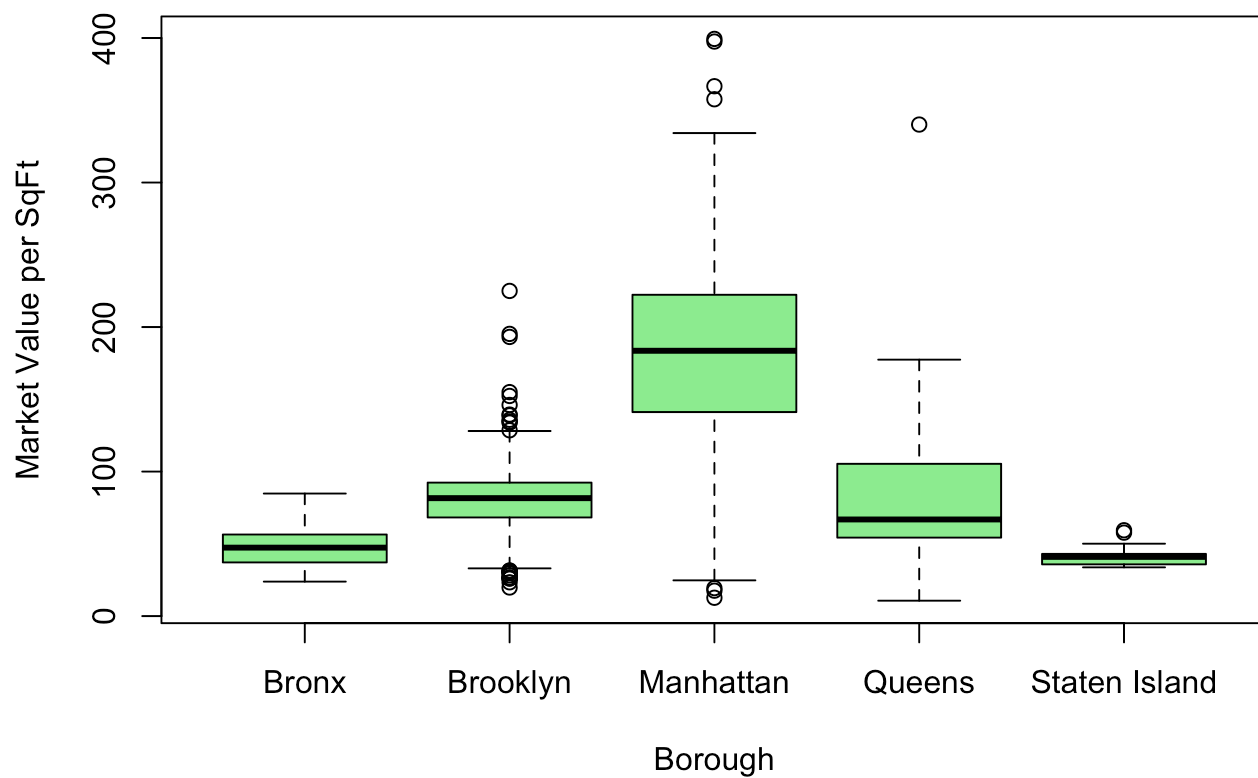
```
# This shows if newer houses are generally more expensive per square foot
plot(housingData$Year.Built, housingData$Market.Value.per.SqFt,
     main = "Market Value vs. Year Built",
     xlab = "Year Built",
     ylab = "Market Value per SqFt",
     pch = 19,      # solid circles
     col = "blue")  # color of points
```

Market Value vs. Year Built



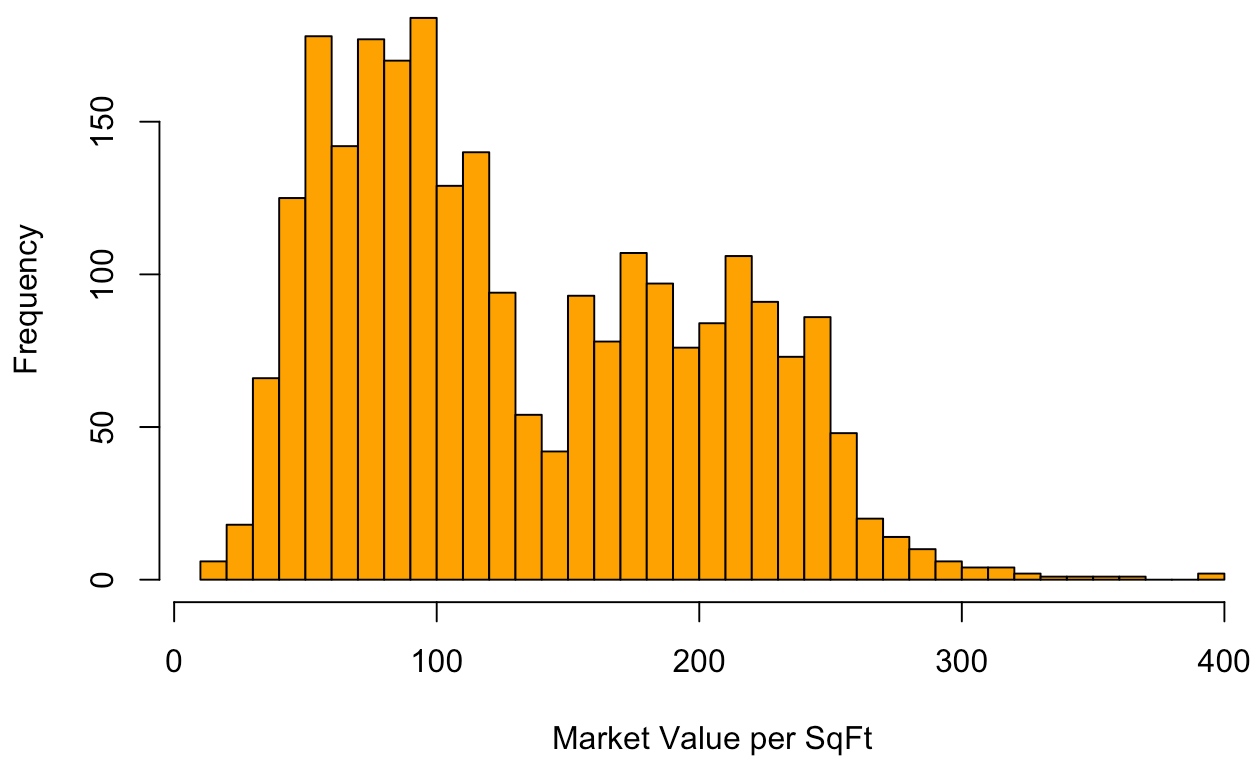
```
boxplot(Market.Value.per.SqFt ~ Boro, data = housingData,
       main = "Market Value per SqFt by Borough",
       xlab = "Borough",
       ylab = "Market Value per SqFt",
       col = "lightgreen")
```

Market Value per SqFt by Borough



```
# This shows how prices are distributed
hist(housingData$Market.Value.per.SqFt,
      main = "Distribution of Market Value per SqFt",
      xlab = "Market Value per SqFt",
      ylab = "Frequency",
      col = "orange",
      breaks = 30) # number of bins
```

Distribution of Market Value per SqFt



This graph shows which boroughs are most/least expensive on average
library(dplyr)

```
boro_avg <- housingData %>%  
  group_by(Boro) %>%  
  summarise(Avg_Value = mean(Market.Value.per.SqFt))
```

```
barplot(boro_avg$Avg_Value,  
        names.arg = boro_avg$Boro,  
        main = "Average Market Value per SqFt by Borough",  
        xlab = "Borough",  
        ylab = "Average Market Value per SqFt",  
        col = "lightblue")
```

Average Market Value per SqFt by Borough



c. Write a summary about your findings from this exercise.

=> Enter your answer here!

In this exercise, we analyzed the housing dataset containing information on neighborhood, borough, market value per square foot, and year built. After cleaning the data, descriptive statistics and graphical visualizations demonstrated that location plays a more critical role in determining property value than building age. While newer constructions often had higher market values, older properties in desirable locations remained competitively priced. The distribution of market values was heavily right-skewed, with a concentration of moderately priced properties and a smaller number of high-value outliers. Significant variability within boroughs further highlighted the diverse nature of local housing markets.